

Hankel-Structured Channel Estimation for Rydberg Atomic Receivers:

Effective-Rank Scaling of the Structural Gain

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Abstract—A Rydberg atomic receiver reads a radio field through an optical transition whose response follows field magnitude, so multi-user channel estimation becomes a phase-retrieval problem in which the noise sits inside the modulus. The established baseline, Gerchberg–Saxton with an EM refinement (EM-GS), treats the channel as unstructured across array elements. We investigate adding a physics-based structural prior: for a uniform linear array the Hankel lifting of each channel column is low rank, so we insert a low-rank Hankel projection into the EM-GS iteration, with the projection order selected from held-out pilots and disabled when the selected order reaches the array’s Hankel rank capacity. In simulation the resulting HS-GS estimator improves on EM-GS by 2.27–3.04 dB at $N = 32$, $P = 30$, but the gain is not universal: it is negative at $N = 8$ and decays to zero as the channel’s normalised effective rank approaches unity. Our results indicate that this normalised effective rank, rather than array size or raw path count, is the coordinate that governs when the prior helps; two frozen cross-generator predictions were met to 0.13 and 0.315 dB. All results are simulation-only.

Index Terms—Rydberg atomic receiver, phase retrieval, channel estimation, Hankel matrix, structured low-rank approximation, effective rank.

I. PROBLEM

A Rydberg atomic receiver detects a radio field through an optical transition whose readout is proportional to field magnitude [1]. With an N -element array serving K single-antenna users over P pilot symbols, the receiver observes

$$\mathbf{Z} = [\mathbf{GS} + \mathbf{B} + \mathbf{W}] \in \mathbb{R}_{\geq 0}^{N \times P}, \quad (1)$$

where $\mathbf{G} \in \mathbb{C}^{N \times K}$ is the unknown channel, $\mathbf{S}$ the pilot matrix, $\mathbf{B}$ a known additive reference field (a local oscillator, needed to break the sign ambiguity of a pure magnitude measurement), and $\mathbf{W}$ circularly symmetric noise. The modulus is applied *after* the noise is added, so the phase of the composite field is lost and (1) cannot be linearised without changing the problem. Channel estimation is therefore biased phase retrieval. The standard treatment alternates a magnitude substitution with a least-squares update — Gerchberg–Saxton [2] — and replaces the hard substitution by the Rician conditional mean, giving EM-GS [3], [4], the baseline throughout this work.

The gap we address. EM-GS estimates $\mathbf{G}$ as $2NK$ free real parameters: nothing in the iteration relates one array element to the next. But a uniform linear array (ULA) illuminated by a finite number of propagation paths produces a channel with

strong deterministic cross-element structure. We investigate whether imposing that structure inside the phase-retrieval iteration improves the estimate, and — the question that occupies most of this summary — *when* it does.

Scope of the contribution. None of the ingredients are new: Hankel low-rank models, Cadzow’s projection [5], structured low-rank approximation [6], effective rank [7], and rank-constrained estimation for atomic receivers [4] all predate this work. To the best of our knowledge the *combination* has not been reported: a Hankel regulariser inside magnitude-only Rydberg EM-GS; order selection that uses no oracle and abstains when the prior cannot be supported; a characterisation of when the prior helps in terms of normalised effective rank; and a prospective test of that characterisation on channel generators not used to build it.

II. HANKEL STRUCTURE AND THE HS-GS ESTIMATOR

A. Why the structure exists

For a ULA with L_k propagation paths, the channel of user k at element n is a sum of complex exponentials in the element index,

$$g_k[n] = \sum_{\ell=1}^{L_k} \alpha_{\ell,k} e^{-j(n-1)\psi_{\ell,k}}, \quad \psi_{\ell,k} = \pi \sin \theta_{\ell,k}. \quad (2)$$

Let $\mathcal{H}(\mathbf{g})$ be the $(N-p) \times (p+1)$ Hankel lifting of the column $\mathbf{g}$. Substituting (2) separates the exponential in the two indices, so each path contributes exactly one rank-one outer product:

$$\mathcal{H}(\mathbf{g}_k) = \sum_{\ell=1}^{L_k} \alpha_{\ell,k} \mathbf{v}_{\ell} \mathbf{w}_{\ell}^{\top} \implies \text{rank } \mathcal{H}(\mathbf{g}_k) \leq L_k. \quad (3)$$

A finite-path ULA channel is thus *exactly* low rank after lifting, while noise and phase-retrieval error are not: they fill the Hankel spectrum. Projecting an EM-GS estimate back toward a low-rank Hankel matrix should therefore remove error the physics forbids while retaining signal it permits. Two consequences shape everything that follows.

The prior has a finite capacity. Since $\mathcal{H}(\mathbf{g})$ is $(N-p) \times (p+1)$, its rank cannot exceed

$$r_{\max}(N) = \max_p \min(N-p, p+1) = \lceil N/2 \rceil. \quad (4)$$

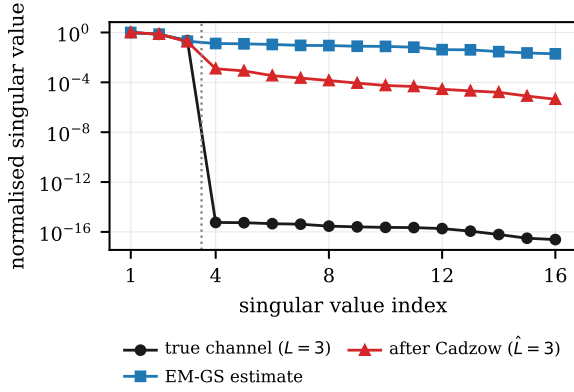


Fig. 1. **Why the prior exists, and what the projection achieves.** [Family EXT, one illustrative realisation.] Normalised Hankel singular values for one channel column, $N = 32$, $L = 3$, $\text{SNR} = 5$ dB. The true channel is exactly rank L ($\sigma_4/\sigma_1 = 5.7 \times 10^{-16}$, the numerical-precision floor). The EM-GS estimate has no such cliff ($\sigma_4/\sigma_1 = 0.130$) — noise fills the spectrum, so the estimate is not a sum of L exponentials. Four Cadzow sweeps restore a cliff at the correct index but only to 1.2×10^{-3} ; the structure is imposed *approximately*, which caps the attainable gain. Single realisation, shown for mechanism, not as a statistical claim.

If $L_k \geq r_{\max}(N)$ the constraint in (3) is vacuous. The value of the prior is therefore bounded by aperture, and this is what makes the method’s benefit conditional rather than universal.

The projection is approximate. The intersection of the fixed-rank matrices and the Hankel matrices is a non-convex variety; Cadzow’s alternating scheme [5] approaches it but the composite map is not an idempotent projection onto it. Fig. 1 shows the residual directly, and it upper-bounds how much structure the method can impose.

B. HS-GS

The proposed estimator inserts the projection into the exact EM-GS iteration:

EM-GS update $\rightarrow$ Hankel lifting $\rightarrow$ rank- $\hat{L}$ Cadzow
projection $\rightarrow$ back to the channel domain $\rightarrow$ next EM-GS
update.

Everything else — initialisation, iteration count, the Rician conditional mean — is unchanged. With the projection disabled, HS-GS reproduces EM-GS bit for bit ($\max |\text{diff}| = 0$, verified in the accompanying test suite), so the projection is provably the only difference between them.

Order selection without an oracle. $\hat{L}$ is never taken from ground truth. A fraction $\nu = 0.3$ of the pilots is held out, each candidate order is scored by the held-out residual, and the best is used for the full run. The *in-sample* residual cannot serve this purpose: constraining $\mathbf{G}$ can only worsen the fit to the pilots used for fitting, so an in-sample criterion always prefers the largest candidate. One scalar $\hat{L}$ is shared by all K users.

Abstention. If the selector returns $\hat{L} \geq r_{\max}(N)$, the projection is the identity and is switched off. This is a designed behaviour, not a failure mode: the estimator declines to impose structure the aperture cannot support, and falls back exactly to EM-GS.

TABLE I
THE THREE EXPERIMENT FAMILIES. DIFFERENT CONFIGURATIONS; NUMBERS FROM ONE ARE NOT COMPARABLE TO NUMBERS FROM ANOTHER AS MEASUREMENTS OF THE SAME ESTIMATOR.

Tag	Store	Configuration
B3	results/track_b/b3/	$N \in \{8, 16, 32\}$, $P \in \{10, 30\}$, 6 SNR pts, 400 trials/pt, $n_{cz} = 4$
EXT	trackB_hankel_emgs/	$P = 30$ only, 7 SNR pts, 600/400/200 trials, $n_{cz} = 4$
TD	results/track_d/partB9/	$N \in \{16, 32, 64\}$, $P = 20$, adaptive order, $n_{cz} = 1$, $\text{SNR} \sim \mathcal{U}[-10, 20]$

III. RESULTS

Table I lists the three experiment families used below. They have different array sizes, pilot counts, SNR grids, sweep counts and pooling rules, and are **never** combined: each number below is tagged with the family it comes from.

A. Does HS-GS improve on EM-GS?

At the principal operating point — $N = 32$, $K = 3$, $P = 30$, 400 paired trials per SNR point — HS-GS improves on EM-GS by +2.27 to +3.04 dB across the tested sweep [B3] (Fig. 2, left).

That gain deserves a comment, because EM-GS is not a weak baseline: against a rank-one Rician bound computed on the same trials, EM-GS tracks the *unconstrained* CRLB to within 0.051 dB for $\text{SNR} \geq 5$ dB [B3]. Nearly 3 dB over an estimator that close to its bound looks contradictory until one notes that the unconstrained bound treats $\mathbf{G}$ as $2NK$ free parameters. Restricting to the geometric manifold of (2) and applying the constrained CRLB [8], [9] gives a bound 7.05–7.11 dB *below* the unconstrained one here [B3]; that gap is the headroom the prior reaches into, and HS-GS recovers part of it. Neither bound formally governs the estimators plotted against it, since all are biased — they are benchmarks, not constraints these curves must satisfy.

B. Aperture: the gain is not universal

Repeating the comparison across array sizes gives a mean gain over the 12 operating points of -0.19 , $+0.78$ and $+2.85$ dB at $N = 8, 16$ and 32 [B3] (Fig. 2, right). **The gain changes sign.**

At $N = 8$ the capacity is $r_{\max} = 4$ by (4) while the simulated path counts satisfy $L_k \in \{3, \dots, 7\}$, so the prior is frequently vacuous or actively harmful: the projection is inactive in 42.5% of trials [B3], and where it is active it often truncates genuine components. The $N = 8$ behaviour is discussed further in Section IV.

This raises the question that organises the rest of the work. Is array size itself the operative variable, or is N a proxy for something else? Capacity (4) suggests the latter: what should matter is not N alone but how much of the available rank capacity the channel actually consumes.

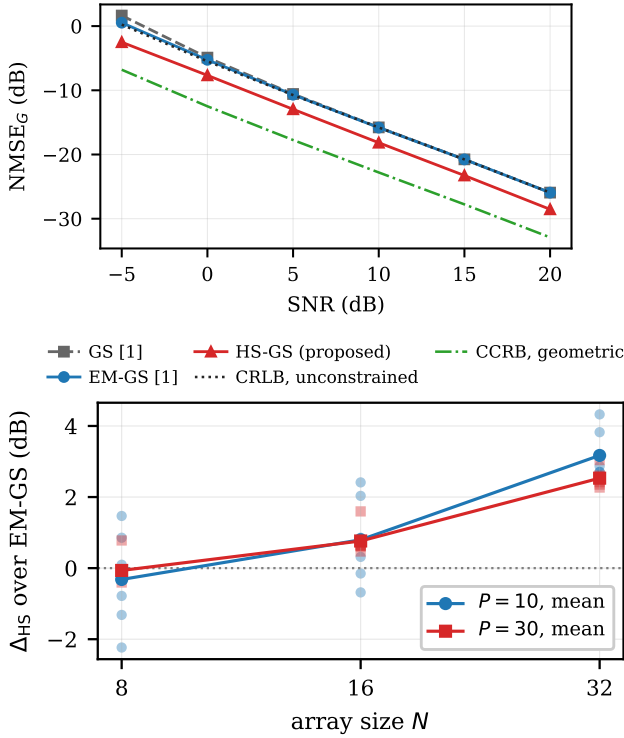


Fig. 2. **The gain is real but conditional.** [Family B3.] *Top:* NMSE versus SNR at $N = 32$, $K = 3$, $P = 30$, $\text{RSR} = 12$ dB, 400 paired trials per point; HS-GS gains 2.27–3.04 dB over EM-GS, and both bounds are plotted for reference (neither governs a biased estimator). *Bottom:* mean gain over the 12 operating points (2 pilot counts $\times$ 6 SNR values) at each array size: -0.19 , $+0.78$, $+2.85$ dB at $N = 8, 16, 32$. **The sign changes with aperture.**

C. A controlled path-count experiment

To separate the two, we fix $N = 32$ and vary only the path count, with L held identical across users and everything else unchanged (Fig. 3). The gain decays monotonically from $+7.04$ dB at $L = 2$ to -0.12 dB at $L = 16 = r_{\max}$ [EXT], reaching a null at $L = 14$ and turning significantly negative at $L = 16$. Over the same sweep EM-GS moves by 0.191 dB in total [EXT] — an estimator that ignores structure should not respond to a change in structure, and it does not, which is the control this experiment needs.

So the operative variable is channel complexity relative to capacity, not aperture. Aperture matters only because it sets the capacity.

D. Effective rank as the coordinate

Raw path count is still an imperfect description of complexity. Two channels with the same L can occupy very different numbers of effective dimensions if their paths are clustered in angle or unequal in strength, since near-collinear or weak components contribute little to the Hankel spectrum. We therefore replace L by the effective rank of Roy and Vetterli [7],

$$r_{\text{eff}} = \exp\left(-\sum_i p_i \ln p_i\right), \quad p_i = \sigma_i / \sum_j \sigma_j, \quad (5)$$

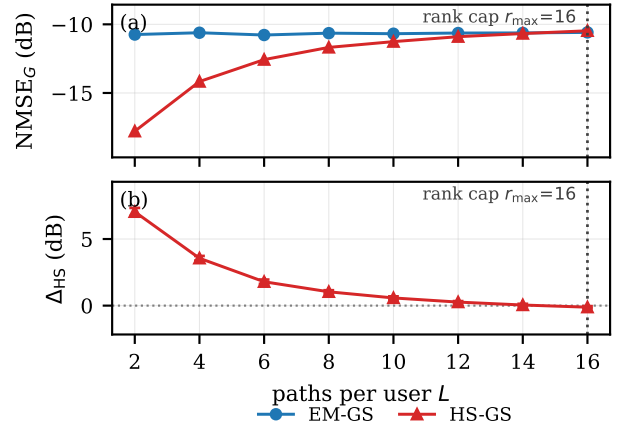


Fig. 3. **Path count is the operative variable, not aperture.** [Family EXT.] Controlled sweep at $N = 32$, $K = 3$, $P = 30$, $\text{SNR} = 5$ dB, 300 trials per point, L fixed and identical across users. (a) EM-GS is flat in L (0.191 dB total) while HS-GS rises to meet it. (b) The paired gain decays from $+7.04$ dB at $L = 2$ to -0.12 dB at $L = 16 = r_{\max}$; the 95% bootstrap band is narrower than the markers. The hypothesis was recorded before the run.

computed on the Hankel lifting, and normalise it by capacity,

$$\rho = r_{\text{eff}} / r_{\max}(N) \in (0, 1]. \quad (6)$$

ρ measures the fraction of the available Hankel rank capacity the channel actually occupies; $\rho \rightarrow 1$ is the regime in which (3) says nothing.

Plotted against ρ , gain curves measured at three different array sizes lie close to one another and cross zero at $\rho = 0.588$, 0.518 and 0.544 for $N = 16$, 32 and 64 [TD] (Fig. 4) — a spread of 0.070 in ρ .

This is an approximate scaling, not a collapse. The *location* of the useful/not-useful boundary is stable across aperture; the gain *magnitude* is not. Comparing $N = 16$ against $N = 64$ over their region of overlap in ρ leaves a residual of mean 0.493 dB and maximum 0.901 dB [TD], and that residual is one-signed rather than scattered, so it is a systematic aperture dependence and not sampling noise. We report ρ as the coordinate that predicts *whether* the prior helps, not one that predicts *how much* it helps to within the measurement error. Two of the three crossings are also extrapolated rather than bracketed by a measured sign change ($N = 32$ is bracketed; $N = 16$ and $N = 64$ are not).

E. Prospective cross-generator validation

A relation fitted and then evaluated on the same data proves little. We therefore froze the ρ -to-gain relation obtained from the ULA experiments above, committed numerical predictions to version control, and only then ran two channel generators that had played no part in constructing it. The relation was *not* refitted afterwards. The prediction rule is a piecewise-linear interpolation on a fixed eight-point table — no regression and no spline — so it can be re-executed exactly.

Generator 1: clustered Saleh–Valenzuela. The clustered multipath model associated with the configuration of Xiao *et al.* [10], which follows the description in [11]. At $\rho =$

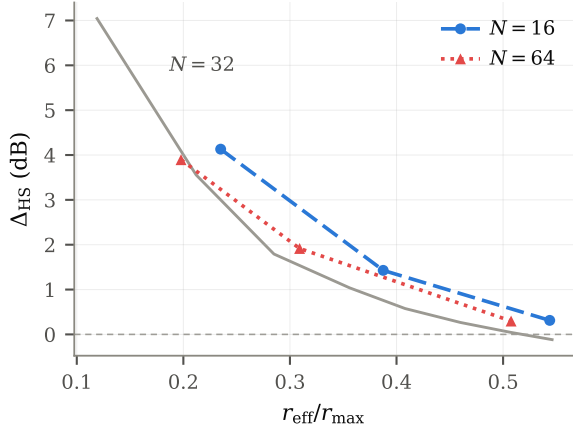


Fig. 4. **Effective rank is the coordinate.** [Family TD.] The same gain plotted against $\rho = r_{\text{eff}}/r_{\text{max}}$ for $N \in \{16, 32, 64\}$, crossing zero at $\rho = 0.588/0.518/0.544$. The boundary location is what is stable across aperture; a one-signed magnitude residual remains (§IV).

0.331 the frozen relation predicted $\Delta_{\text{HS}} = +1.30$ dB; the measurement over 276 trials gave $+1.173$ dB with 95% CI $[+1.017, +1.380]$ — an error of 0.13 dB [TD].

Generator 2: the $L = 10$ configuration. The simulation configuration reported in [12]. At $\rho = 0.400$ the prediction was $+0.648$ dB with a stated interval $[+0.367, +1.015]$; the measurement over 266 trials gave $+0.963$ dB with 95% CI $[+0.808, +1.086]$ — an error of 0.315 dB, inside the stated prediction interval but on its upper edge [TD].

What this does and does not show. These are *cross-generator* or distribution-shift tests, not out-of-model tests: both generators still produce channels of the form (2), and only the distribution over $(L_k, \alpha_{\ell,k}, \theta_{\ell,k})$ changes. So this is evidence that ρ transfers across path distributions, not that it survives a violation of the ULA sum-of-exponentials model. Within that scope, a frozen coordinate predicted a structural gain on unseen channel statistics to 0.13 and 0.315 dB, and the weaker second case is reported alongside the first.

IV. LIMITATIONS

These define the regime in which the method is useful and are reported in full.

$N = 8$ is a mixed result, not a small positive one. The gain is $+1.47$ dB at -5 dB SNR ($P = 10$) but falls to -2.23 dB at high SNR [B3], because truncation bias does not shrink with noise: as SNR rises the variance term falls while the bias term does not, so the trade stops paying and then reverses. The sign of any single scalar summary at $N = 8$ depends on the pooling rule, which is why per-point values are reported. **The effective-rank scaling is approximate.** The boundary location is stable across aperture; the gain magnitude is not, leaving an unexplained one-signed residual of mean 0.493 dB (maximum 0.901 dB) between $N = 16$ and $N = 64$ [TD]. **A pre-registered criterion narrowly failed.** A K -invariance test at fixed pilot adequacy $P/2K = 3.33$ passes comfortably at $\text{SNR} \geq 5$ dB (spread 0.094 dB) but, pooled

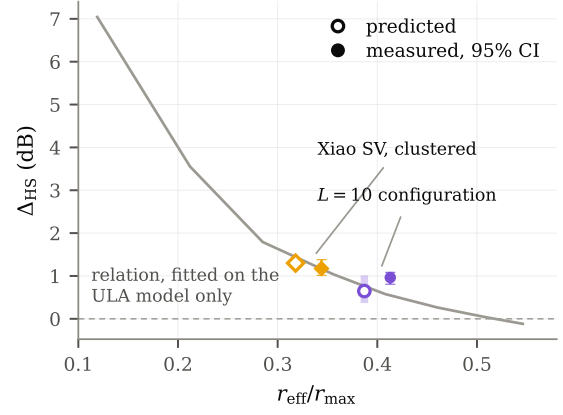


Fig. 5. **Predictions registered before measurement.** [Family TD, $N = 32$, $K = 3$, $P = 20$, adaptive order, $n_{\text{cz}} = 1$, $\text{SNR} \sim \mathcal{U}[-10, 20]$ dB.] Hollow markers are what the frozen relation predicted; filled markers with bars are what was measured, with 95% confidence intervals. The grey curve is the relation itself, fitted on the ULA experiments only and not refitted for either generator. The two are drawn at a small horizontal offset because at the same abscissa they overlap — which is the result, but would hide which is which. Errors: 0.13 dB (clustered Saleh–Valenzuela, 276 trials) and 0.315 dB ($L = 10$ config., 266 trials).

over all SNR, gives 0.294 dB against a falsifier registered at 0.3 dB — missed by 0.006 dB [TD]; reported rather than dropped. **Simulation only.** No measured hardware and no recorded channel data; the atomic response enters only through (1), and all channels are geometric ULA channels of the form (2). **Order selection is expensive,** accounting for 57.7–85.5% of HS-GS runtime [B3]; a coarse-to-fine search or reuse of $\hat{L}$ across coherence blocks is the obvious target, and neither was attempted. **Neither bound governs the estimators compared to them,** as noted in §III-A.

V. CONCLUSION

Hankel structural regularisation can substantially improve magnitude-only Rydberg channel estimation, but only when the channel occupies a sufficiently small fraction of the array’s Hankel rank capacity: we observe $+2.27$ to $+3.04$ dB at $N = 32$, $P = 30$ [B3], and a sign reversal at $N = 8$. The more general finding is that the normalised effective rank $\rho = r_{\text{eff}}/r_{\text{max}}$ is more informative than raw path count or array size alone in describing when the structural prior is useful — a controlled sweep at fixed N drives the gain from $+7.04$ to -0.12 dB as L approaches capacity [EXT], and the zero crossing in ρ sits within a spread of 0.070 across three array sizes [TD]. Two frozen cross-generator predictions were met to 0.13 and 0.315 dB, which is preliminary predictive evidence rather than a validated scaling law: the magnitude residual across aperture is unexplained, one pre-registered criterion narrowly failed, and every result here is simulation-only on geometric ULA channels. Broader channel families, non-idealities of a physical vapour cell, and measurements on hardware are all required before the coordinate can be claimed to generalise.

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